

Divisible perturbations of the weight:

$$Z(a + u^k \tilde{J}, \eta_0 + u^k \tilde{\eta}) \sim \eta_0 C n^{-p/2} (\log n)^m$$

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2026-09-06

Abstract

The mechanism of unit 65 extends to the weight. For $\eta = \eta_0 + u^k \tilde{\eta}$ and $\xi = a + u^k \tilde{J}$ with $\tilde{\eta}, \tilde{J}$ continuous on the box, the weight perturbation carries the extra factor $u^k = t/\sqrt{n}$, and $te^{-(\beta/4)t^2} \leq 4/\beta + 1$ gives $|Z(\xi, \eta) - \eta_0 Z(\xi, 1)| \leq \frac{L_\eta(4/\beta+1)}{\sqrt{n}} Z_{\beta/2, 2a}(n)$ for $n \geq 16L^2$. Combined with unit 65, $Z(\xi, \eta) - \eta_0 Z(a, 1) = O(n^{-1/2} Z_{\beta/2, 2a})$, so for $\eta_0 \neq 0$ the standard integral is $\sim \eta_0 C n^{-p/2} (\log n)^m$ with the constant- ξ exponent, multiplicity and constant: the leading coefficient is $\eta(0)$ times the population constant, as the paper's remark on the sign of η anticipates. Zero sorry/axiom.

1 Setting

Both perturbations vanish on the exceptional divisor to order u^k , which is exactly what makes them lower order: after the substitution $t = \sqrt{n} u^k$ each contributes a factor $t/\sqrt{n}$ or $t^2/\sqrt{n}$, absorbed by comparing with the kernel at the halved temperature.

2 The things formalised

```

noncomputable def boxIntegralXiEta ( a b c : ) {d : } (k h : Fin d → )
  (J : (Fin d → ) → ) : := u, ( i, u i ^ h i ) * u
  * Real.exp ( - * (c * i, u i ^ k i) ^ 2 + * (c * i, u i ^ k i) * (a + J u) )
  (boxMeasure b d)
theorem mul_exp_neg_quarter_le ( t : ) (h : 0 < ) :
  t * Real.exp (-( / 4) * t ^ 2) 4 / + 1
theorem boxIntegralXiEta_sub_le ... (hnL : 16 * L ^ 2 n) :
  |boxIntegralXiEta a b (Real.sqrt n) k h (fun u => ( i, u i ^ k i) * J u)
    (fun u => + ( i, u i ^ k i) * u)
  - * boxIntegralXi a b (Real.sqrt n) k h (fun u => ( i, u i ^ k i) * J u)|
  L * (4 / + 1) / Real.sqrt n * boxIntegralFin ( / 2) (2 * a) b (Real.sqrt n) k h
theorem boxIntegralXiEta_isEquivalent ... (h : 0) :
  (p : ) (m : ) (C : ), 0 < C ( i, p finExp k h i)
  m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
  (fun n => boxIntegralXiEta a b (Real.sqrt n) k h (fun u => ( i, u i ^ k i) * J u)
    (fun u => + ( i, u i ^ k i) * u))
  ~[atTop] fun n => * C * (n ^ (-(p / 2))) * Real.log n ^ m

```

3 Proof strategy

Pointwise, the weight-perturbation integrand is $u^h(t/\sqrt{n})\tilde{\eta}e^{-\beta t^2+\beta t(a+t\tilde{J}/\sqrt{n})}$; the exponential is at most $e^{-(\beta/4)t^2}e^{-(\beta/2)t^2+\beta ta}$ when $L/\sqrt{n} \leq 1/4$, and $te^{-(\beta/4)t^2} \leq 4/\beta + 1$ follows from $t \leq t^2 + 1$ and $t^2 \leq \frac{4}{\beta}e^{(\beta/4)t^2}$ (`Real.add_one_le_exp`). Integration as in unit 65. For the asymptotic the two differences are combined by the triangle inequality into an $O(n^{-1/2}Z_{\beta/2,2a})$ bound, which is $o(\eta_0 Z(a))$; `IsEquivalent.add_isLittle0` concludes after multiplying unit 59's equivalence by the constant η_0 .

4 Roadblocks and resolutions

- Rewriting t^2 as $\frac{4}{\beta}(\frac{\beta}{4}t^2)$ with `rw` also rewrote inside the exponential; a `calc` step instead.
- Dot notation on an `IsEquivalent` fact fell through to `IsLittle0.const_mul_left` (the definition unfolds); multiply by a constant via `IsEquivalent.refl.mul` and `congr`.
- `rw [one_div]` hit the exponent $1/2$ before the intended $1/\sqrt{n}$; use `(one_div _).symm` as a term.

5 What was Mathlib, what was new

Mathlib: `Real.add_one_le_exp`, `Real.one_le_exp`, `abs_add_le`, `IsEquivalent.add_isLittle0`, `IsLittle0.mul_isBig0`. New: the weight-perturbation bound and the joint theorem for divisible ξ and η perturbations.

6 Lessons

Once one perturbation is handled by comparison with a halved-temperature kernel, further perturbations of the same divisibility type follow the same template; the triangle inequality assembles them.

7 Outcome

The multiplicity theorem now covers weights and fluctuation functions of the form $\eta_0 + u^k\tilde{\eta}$, $a + u^k\tilde{J}$, with leading coefficient η_0 times the population constant. What remains for general analytic ξ, η is the genuine leading-coefficient dependence on their restrictions to the minimal divisor components.